

# The radix-2 FFT:

Notice that

$$x[k + \frac{N}{2}] = G[k + \frac{N}{2}] + W_N^{k + \frac{N}{2}} H[k + \frac{N}{2}]$$

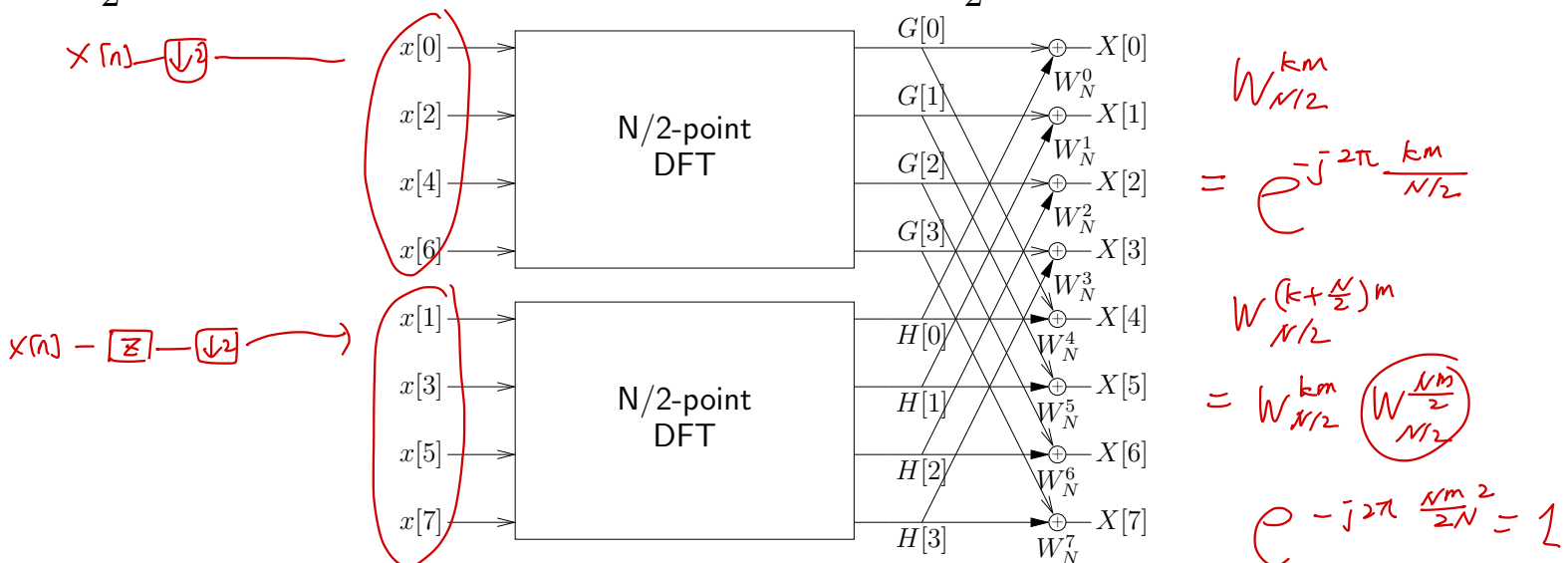
$$\begin{aligned} X[k] &= \sum_{n=0}^{N-1} x[n] W_N^{kn} \quad \text{for } W_N^q \triangleq e^{-j \frac{2\pi}{N} q} \\ &= \sum_{n \text{ even}} x[n] W_N^{kn} + \sum_{n \text{ odd}} x[n] W_N^{kn} \\ &= \sum_{m=0}^{N/2-1} x[2m] W_N^{k2m} + \sum_{m=0}^{N/2-1} x[2m+1] W_N^{k(2m+1)} \end{aligned}$$

$W_N^{k(2m+1)} = W_N^{k2m} W_N^k = W_N^{km} W_N^k$

Because  $W_N^{2q} = W_{N/2}^q$  and  $W_N^{q+p} = W_N^q W_N^p$ , we can write

$$\begin{aligned} X[k] &= \underbrace{\sum_{m=0}^{N/2-1} x[2m] W_{N/2}^{km}}_{\triangleq G[k]} + W_N^k \underbrace{\sum_{m=0}^{N/2-1} x[2m+1] W_{N/2}^{km}}_{\triangleq H[k]} \end{aligned}$$

Note:  $G[k]$  is an  $\frac{N}{2}$ -DFT of the even samples, and  $H[k]$  is an  $\frac{N}{2}$ -DFT of the odd samples. Both are  $\frac{N}{2}$ -periodic in  $k$ .

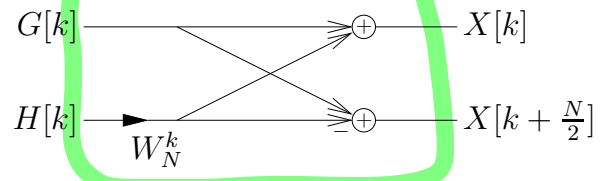
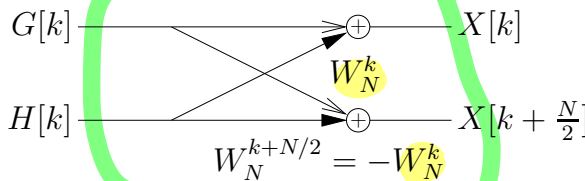


$$W_N^{k+N/2} = W_N^k W_N^{N/2}$$

$$W_N^{N/2} = e^{-j2\pi \frac{N/2}{N}} = e^{-j\pi} = -1$$

## The radix-2 FFT (cont.):

- The final processing stage can be simplified from  $N$  mults to  $\frac{N}{2}$  mults because the following are equivalent:

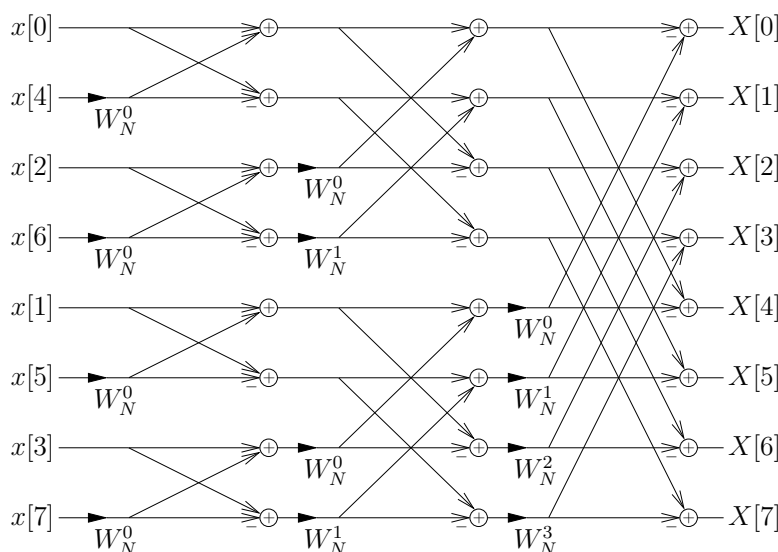


- Let's examine the complexity of this split-in-half method:

- direct  $N$ -DFT:  $N^2$  mults,
- split-in-half:  $(\frac{N}{2})^2 + (\frac{N}{2})^2 + \frac{N}{2} = \frac{1}{2}(N^2 + N)$  mults,

Since  $\frac{1}{2}(N^2 + N) < N^2$  for any  $N > 1$ , *it pays to split!*

- Thus, we split each  $\frac{N}{2}$ -DFT into two  $\frac{N}{4}$ -DFTs (combining their outputs using twiddle factors), repeat with each  $\frac{N}{4}$ -DFT, and so on, finally obtaining



In total,  $(\log_2 N)$  stages  $\times \frac{N}{2}$  mults/stage  $= \frac{N}{2} \log_2 N$  mults.

## Fast convolution:

The FFT gives us a very fast way to do linear convolution.

- Say that we want to *linearly* convolve an  $M$ -length signal  $x[n]$  with an  $L$ -length signal  $h[n]$ , i.e.,

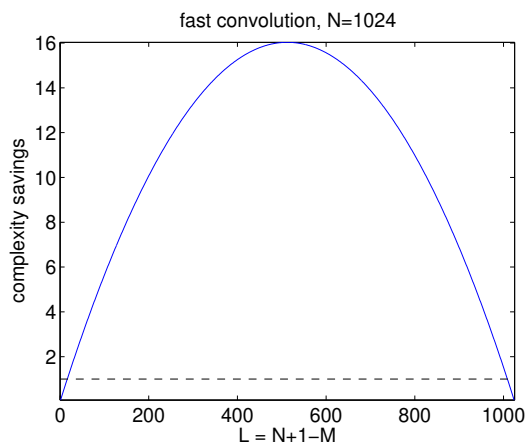
$$y[n] = \sum_{l=0}^{L-1} h[l]x[n-l],$$

requiring  $\approx L$  mults per output, or  $ML$  mults in total.

- If we zero-pad both sequences to length  $N \geq M + L - 1$ , then  $N$ -circular convolution yields the same outputs as linear convolution. (Recall page 11.)
- Furthermore,  $N$ -circular convolution can be implemented as multiplication in the  $N$ -DFT domain:

$$x[n] \circledast h[n] = \mathcal{F}_{N\text{-DFT}}^{-1} \left\{ \mathcal{F}_{N\text{-DFT}} \{x[n]\} \cdot \mathcal{F}_{N\text{-DFT}} \{h[n]\} \right\}$$

and, if we use FFT/IFFT to implement the DFT/IDFT, this requires only  $3 \times \frac{N}{2} \log_2 N + N$  total mults.



Much cheaper  
unless  $M$  or  $L$   
very small!

## Overlap-Save:

- Say that we want to filter  $M$ -length signal  $x[n]$  with  $L$ -length filter impulse response  $h[n]$ .
- Say that we have access to an  $N$ -length FFT algorithm, where  $L < N < M + L - 1$ .

*Rather than zero-padding  $x[n]$ , we break it up into overlapping segments of length  $N$ , and perform fast convolution on each one.*

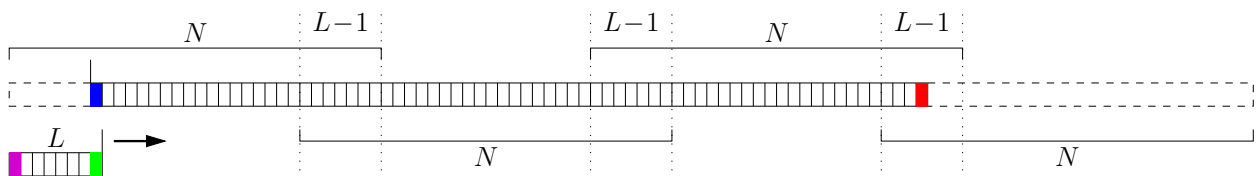
- Key quantities:

$M + L - 1 =$  # linear-convolution outputs to compute

$N - L + 1 =$  # linear-conv outputs per  $N$ -circular conv

$\lceil \frac{M+L-1}{N-L+1} \rceil =$  #  $N$ -circular convolutions needed in total

Example with  $L = 8$ ,  $N = 32$ ,  $M = 72$ :



- Total multiplies required for overlap-save:

$$\underbrace{\frac{N}{2} \log_2 N}_{H=\text{fft}(h)} + \lceil \frac{M+L-1}{N-L+1} \rceil \left( \underbrace{\frac{N}{2} \log_2 N}_{X=\text{fft}(x)} + \underbrace{N}_{Y=X.*H} + \underbrace{\frac{N}{2} \log_2 N}_{y=\text{ifft}(Y)} \right)$$

Compare to  $ML$  multiplies for standard linear conv.

## Overlap-Save Example:

$a \ b \ c \ d \ 0$   
 $/ \ / \ 0 \ 0 \ 0$

- Say we want to compute the linear convolution  $[a, b, c, d] * [1, 1]$  using several 3-circular convolutions.
- We know that the desired output is

$$[a, a+b, b+c, c+d, d]$$

- In this example,  $M=4$ ,  $L=2$ , and  $N=3$ , which means:
  - $M+L-1 = 5$  linear convolution outputs
  - $N-L+1 = 2$  good outputs per  $N$ -circular convolution
  - $\lceil \frac{M+L-1}{N-L+1} \rceil = 3$   $N$ -circular convolutions
- We prepare the signal by inserting  $L-1 = 1$  zeros in front and as many as needed in back.

$$\overbrace{0, a, b} \quad \overbrace{c, d, 0, 0}$$

The  $N$ -blocks overlap by  $L-1 = 1$  samples, as shown.

- The  $N$ -circular convolutions manifest as:

$$[0, a, b] \textcircled{3} [1, 1] = [\cancel{b}, a, a+b]$$

$$[b, c, d] \textcircled{3} [1, 1] = [\cancel{d+\cancel{b}}, b+c, c+d]$$

$$[d, 0, 0] \textcircled{3} [1, 1] = [\cancel{d}, d, 0]$$

and we discard the first  $L-1 = 1$  samples of each.

- Concatenating the good circular-convolution outputs gives

$$[a, a+b, b+c, c+d, d, 0]$$

which is the desired result (plus some zero padding).

$$\begin{aligned}\underline{X} &= W \underline{x} & \|\underline{X}\|^2 &= \underline{X}^H \underline{X} = (W \underline{x})^H W \underline{x} \\ & & &= \underline{x}^H W^H W \underline{x} = N \underline{x}^H \underline{x} = N \|\underline{x}\|^2\end{aligned}$$

## Matrix/vector formulations:

Many of the properties we've seen so far can be efficiently described using matrices and vectors.

- The  $N$ -DFT  $X[k] = \sum_{n=0}^{N-1} x[n] e^{-j \frac{2\pi}{N} kn}$  can be written in matrix/vector form as

$$\underbrace{\begin{bmatrix} X[0] \\ X[1] \\ \vdots \\ X[N-1] \end{bmatrix}}_{\mathbf{X}} = \underbrace{\begin{bmatrix} e^{-j \frac{2\pi}{N} \cdot 0 \cdot 0} & e^{-j \frac{2\pi}{N} \cdot 0 \cdot 1} & \dots & e^{-j \frac{2\pi}{N} \cdot 0 \cdot (N-1)} \\ e^{-j \frac{2\pi}{N} \cdot 1 \cdot 0} & e^{-j \frac{2\pi}{N} \cdot 1 \cdot 1} & \dots & e^{-j \frac{2\pi}{N} \cdot 1 \cdot (N-1)} \\ \vdots & \vdots & \ddots & \vdots \\ e^{-j \frac{2\pi}{N} \cdot (N-1) \cdot 0} & e^{-j \frac{2\pi}{N} \cdot (N-1) \cdot 1} & \dots & e^{-j \frac{2\pi}{N} \cdot (N-1) \cdot (N-1)} \end{bmatrix}}_{\mathbf{W}} \underbrace{\begin{bmatrix} x[0] \\ x[1] \\ \vdots \\ x[N-1] \end{bmatrix}}_{\mathbf{x}}.$$

The  $k^{th}$  row &  $n^{th}$  column of DFT matrix  $\mathbf{W}$  is  $e^{-j \frac{2\pi}{N} kn}$ .

- The IDFT matrix is  $\mathbf{W}^{-1} = \frac{1}{N} \mathbf{W}^H$ , where  $(\cdot)^H$  denotes the "Hermitian," i.e., conjugate transpose. Notice

$$\frac{1}{N} \mathbf{W}^H \mathbf{W} = \mathbf{I} = \frac{1}{N} \mathbf{W} \mathbf{W}^H,$$

since

$$\begin{aligned}\left[\frac{1}{N} \mathbf{W} \mathbf{W}^H\right]_{k,l} &= \frac{1}{N} \sum_{n=0}^{N-1} e^{-j \frac{2\pi}{N} kn} e^{j \frac{2\pi}{N} ln} \\ &= \frac{1}{N} \sum_{n=0}^{N-1} e^{-j \frac{2\pi}{N} n(k-l)} \\ &= \begin{cases} 1 & k = l \\ 0 & k \neq l \end{cases} = [\mathbf{I}]_{k,l}.\end{aligned}$$

## Matrix/vector formulations (cont.):

- An important class of matrices are “unitary” matrices, i.e., matrices  $U$  such that

$$U^H U = I = U U^H.$$

Note that the normalized DFT matrix  $\frac{1}{\sqrt{N}} \mathbf{W}$  is unitary.

- Unitary matrices implement *rotations*:  $U\mathbf{x}$  is a rotation of  $\mathbf{x} \in \mathbb{C}^N$  in the space  $\mathbb{C}^N$ . Essentially, the (scaled) DFT  $\frac{1}{\sqrt{N}} \mathbf{W}$  rotates the time-domain signal vector  $\mathbf{x}$  so that its Fourier components (i.e., complex exponentials) are visible. Sometimes, this is a better “view” of the signal.
- Notice that rotating a vector does not change its length, as measured by the standard Euclidean norm

$$\|\mathbf{x}\|_2 \triangleq \sqrt{\sum_{n=0}^{N-1} |x[n]|^2}.$$

In other words,  $\|U\mathbf{x}\|_2 = \|\mathbf{x}\|_2$  for any unitary  $U$ . This explains Parseval’s theorem for the DFT:

$$\sum_{n=0}^{N-1} |x[n]|^2 = \|\mathbf{x}\|_2^2 = \underbrace{\left\| \frac{1}{\sqrt{N}} \mathbf{W} \mathbf{x} \right\|_2^2}_U = \frac{1}{N} \|\mathbf{W} \mathbf{x}\|_2^2 = \frac{1}{N} \sum_{k=0}^{N-1} |X[k]|^2.$$

## Matrix/vector formulations (cont.):

- Linear convolution  $y[n] = \sum_{l=-\infty}^{\infty} x[l]h[n-l]$  can be written as, for example,

$$\begin{bmatrix} y[0] \\ y[1] \\ y[2] \\ y[3] \\ y[4] \\ y[5] \\ y[6] \end{bmatrix} = \begin{bmatrix} h[0] & 0 & 0 & 0 \\ h[1] & h[0] & 0 & 0 \\ h[2] & h[1] & h[0] & 0 \\ h[3] & h[2] & h[1] & h[0] \\ 0 & h[3] & h[2] & h[1] \\ 0 & 0 & h[3] & h[2] \\ 0 & 0 & 0 & h[3] \end{bmatrix} \begin{bmatrix} x[0] \\ x[1] \\ x[2] \\ x[3] \end{bmatrix},$$

while  $N$ -circular convolution  $y[n] = \sum_{l=0}^{N-1} x[l]h[\langle n-l \rangle_N]$  can be written as

$$\begin{bmatrix} y[0] \\ y[1] \\ y[2] \\ y[3] \end{bmatrix} = \begin{bmatrix} h[0] & h[3] & h[2] & h[1] \\ h[1] & h[0] & h[3] & h[2] \\ h[2] & h[1] & h[0] & h[3] \\ h[3] & h[2] & h[1] & h[0] \end{bmatrix} \begin{bmatrix} x[0] \\ x[1] \\ x[2] \\ x[3] \end{bmatrix},$$

where the matrix has a “circulant” structure.

- Circulant  $\mathbf{H}$  with first column  $\mathbf{h}$  has eigendecomposition

$$\mathbf{H} = \frac{1}{N} \mathbf{W}^H \mathbf{D} \mathbf{W} \quad \text{where} \quad \mathbf{D} = \text{diag}(\mathbf{W} \mathbf{h}).$$

Thus, circular convolution  $\mathbf{y} = \mathbf{H} \mathbf{x}$  can be written as

$$\mathbf{y} = \frac{1}{N} \mathbf{W}^H \mathbf{D} \mathbf{W} \mathbf{x}$$

which can be recognized as DFT-domain multiplication!